NGUYEN DINH HUY

Mechatronics & Robotics Engineer

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SUMMARY

Mechatronics graduate from HCMUT with a foundation in mechanical design, electrical-control systems, and system integration. Capable of 3D modeling, technical drafting, component selection, and building operating principles for machinery. Completed academic and personal projects. Seeking to develop expertise in R&D Mechanical Design, Automation, and Robotics/AMR Integration.

EDUCATION

Ho Chi Minh City University of Technology (HCMUT) **2022 – 2026**
Bachelor of Engineering in Mechatronics **GPA: 3.0/4.0**

- **Graduation Thesis:** DESIGN OF AN UNDER-RIDE AUTONOMOUS MOBILE ROBOT FOR RACK LIFTING.
- **Key Coursework:** Microcontrollers, Hydraulics & Pneumatics, Robotics Engineering, PLC Systems in Robotics & Automation, Servo System Control, Dynamics & Control, Manufacturing Processes.
- **Language Certificate:** TOEIC 650.

TECHNICAL PROJECTS

Under-Ride Autonomous Mobile Robot for Rack Lifting (Graduation Thesis) **03/2026 – 09/2026**
Role: Full-System Design & Software Development

- Designed and 3D-modeled the mechanical structure and lifting mechanism.
- Developed the electrical-control architecture; calculated and selected mechatronic components.
- Configured robot navigation behavior (Nav2) and developed the under-ride docking program in ROS 2; fully verified the system via Gazebo simulation.

Hexapod Robot **09/2025 – 12/2025**
Role: Mechanical Design & Control Programming

- Designed the mechanical structure, performed structural analysis, and built 3D models; calculated and selected appropriate motors.
- Programmed locomotion gaits for the 3D-printed physical model using a microcontroller.

Color-Sorting Line-Following Robot **02/2025 – 06/2025**
Role: Simulation & Control Programming

- Calculated differential kinematics, identified DC motor transfer functions, and designed a PID controller for the system.
- Developed color-based sorting algorithms, simulated and programmed the line-following robot using ESP32.

Other Projects: Gearbox Design Project, 4-DOF Robotic Arm Simulation and Control, Stepper Motor Control.

WORK EXPERIENCE

Tien Phat Automation Co., Ltd **06/2024 – 08/2024**
Role: Intern

- Learned 3D modeling and exported mechanical drawings; supported design, machining, and assembly processes.

Minh Tan Mechanical Production - Trading - Service Co., Ltd **06/2025 – 09/2025**
Role: Intern

- Extracted technical drawings, operated CNC machines, and inspected product dimensions.
- Supported assembly, maintenance, and machine repair at factory/industrial park sites.

KNOWLEDGE AND SKILLS

Professional Knowledge:

- **Mechanical Design:** 3D modeling, technical and assembly drafting, mechanism design, basic mechanical calculations, material selection, and structural analysis.
- **Electrical & Electronics:** Designing and implementing schematics and wiring diagrams; selecting electrical components, motors, and sensors.
- **Robotics & Automation:** Basic knowledge of MCUs, PLCs, and various types of motors.
- **Programming & Control:** Reading and modifying microcontroller code, basic PID tuning, configuring, testing, and calibrating robot control systems.

Skills & Tools:

- **CAD Design:** SolidWorks, AutoCAD Mechanical.
- **Simulation & Electrical:** MATLAB, Proteus, AutoCAD Electrical.
- **Programming & Robotics:** Arduino IDE, Visual Studio Code (VS Code), Ubuntu, ROS 2, Gazebo.